RaceShowdown Project Notes

Project Overview

RaceShowdown compares two athletes in a toy race-eating scenario:

- both athletes first eat the same number of hot dogs,
- both athletes then run the same distance,
- the faster total time wins.

The codebase evaluates this matchup in two ways:

- a **deterministic model**, which compares mean total times,
- a **probabilistic model**, which adds noise and estimates win probabilities by Monte Carlo simulation.

The main entry point loads the TOML configuration, builds a distance-hot-dog grid, and produces:

- a probabilistic phase diagram, showing $P(\text{athlete A wins})$,
- a deterministic phase diagram, showing which athlete has the lower mean time,
- printed reference probabilities at a few named scenarios from the config file.

Current Mathematical Model

Inputs

For a race we specify:

- $n \geq 0$: number of hot dogs eaten,
- $d > 0$: running distance in meters.

All distances are in meters and all times are in seconds.

Eating Time

For athlete i, the time to eat the m-th hot dog is modeled as

$$e_i(m) = a_i + b_i m,$$

where:

- a_i is the base eating time,
- b_i is the per-hot-dog slowdown.

The cumulative mean time to eat n hot dogs is

$$E_i(n) = \sum_{m=1}^{n} (a_i + b_i m).$$

Post-Eating Running Penalty

After eating n hot dogs, the running component is multiplied by

$$M_i(n) = 1 + c_i n + q_i n^2,$$

where:

- c_i is a linear post-eating penalty,
- q_i is a quadratic post-eating penalty.

This is a simple way to capture the idea that running becomes harder after heavier eating.

Running Time: Two-Regime Model

The current project does *not* use a single running curve for all distances. Instead, it uses a sprint regime up to a threshold distance d_s and an endurance regime beyond that threshold.

Why the old single-curve idea was not enough. A single Riegel-style power law can fit middle and long distances reasonably well, but it is too rigid for sprint specialists. If one exponent must explain both 100m and 5000m, then an athlete who is exceptional at short sprints tends to become unrealistically dominant at longer distances too. The current model separates these effects.

Endurance regime. For $d > d_s$, athlete i uses a Riegel-style law

$$R_i^{\text{long}}(d) = T_{i,\text{ref}} \left(\frac{d}{d_{\text{ref}}} \right)^{k_i},$$

where:

- $T_{i,\text{ref}}$ is the runner's time at the endurance reference distance,
- d_{ref} is the endurance reference distance,
- k_i is the endurance exponent.

Interpretation:

- lower $T_{i,\text{ref}}$ means a faster runner around the reference distance,
- lower k_i means better endurance,
- larger k_i means the runner slows down faster as distance increases.

Sprint regime. For $d \leq d_s$, athlete i uses a sprint-oriented power law

$$R_i^{\text{short}}(d) = T_{i,\text{sprint}} \left(\frac{d}{d_{\text{sprint}}} \right)^{\alpha_i},$$

where:

- $T_{i,\text{sprint}}$ is the athlete's sprint reference time,
- d_{sprint} is the sprint reference distance,
- α_i is chosen so that the short and long regimes match exactly at d_s.

The continuity condition is

$$R_i^{\text{short}}(d_s) = R_i^{\text{long}}(d_s),$$

which gives

$$\alpha_i = \frac{\log\left(R_i^{\text{long}}(d_s)/T_{i,\text{sprint}}\right)}{\log(d_s/d_{\text{sprint}})}.$$

This means:

- **sprint ability** is primarily controlled by $T_{i,\text{sprint}}$,
- **endurance ability** is primarily controlled by $T_{i,\text{ref}}$ and k_i,
- the transition at d_s is continuous by construction.

Full running model. The mean running time used by the simulator is

$$R_i(d,n) = \begin{cases} R_i^{\text{short}}(d)\, M_i(n), & d \le d_s, \\ R_i^{\text{long}}(d)\, M_i(n), & d > d_s. \end{cases}$$

Total Mean Race Time

The full deterministic race time is

$$T_i(d,n) = E_i(n) + R_i(d,n).$$

The deterministic comparison between athletes A and B is

$$\Delta(d,n) = T_A(d,n) - T_B(d,n).$$

Interpretation:

- $\Delta(d,n) < 0$: athlete A is faster on mean time,
- $\Delta(d,n) > 0$: athlete B is faster on mean time,
- $\Delta(d,n) = 0$: deterministic tie.

Probabilistic Model

The Monte Carlo layer samples noise around the mean model:

- additive eating noise,
- multiplicative sprint-reference-time noise,
- multiplicative endurance-reference-time noise,
- multiplicative endurance-exponent noise,
- multiplicative post-eating-penalty noise.

This produces random total race times $\widetilde{T}_A(d,n)$ and $\widetilde{T}_B(d,n)$. The estimated win probability is

$$p(d,n) = \Pr\big(\widetilde{T}_A(d,n) < \widetilde{T}_B(d,n)\big).$$

Configuration Guide

The configuration file is `config/config.toml`. It has three main sections:

- `[simulation]`
- `[running_model]`
- `[running_model.runners.<name>]`

Simulation Attributes

`athlete_a`, `athlete_b` The two named athletes compared in the plots and reference outputs.

`min_distance_m`, `max_distance_m` The distance range used for the phase diagrams.

`distance_samples` Number of distance grid points between the minimum and maximum.

`max_hotdogs` Largest hot-dog count shown on the grid.

`monte_carlo_samples` Number of Monte Carlo samples per grid cell for the probabilistic plot.

`random_seed` Seed used for reproducible simulations.

`[[simulation.reference_checks]]` A small list of named scenarios printed after the plots as quick sanity checks.

Current choices. The project currently studies distances from 100m to 5000m, uses 120 distance samples, and considers 0 to 20 hot dogs. This range matches the intended scope of the running model: short sprint through 5k, not 10k or marathon modeling.

Global Running-Model Attributes

`sprint_threshold_distance` The regime switch distance d_s.

`sprint_reference_distance` The short-distance anchor d_{sprint} used with `sprint_ref_time`.

`reference_distance` The endurance reference distance d_{ref} used with `ref_time`.

`max_modeled_distance` Hard upper bound for the supported running model.

Current choices.

- `sprint_threshold_distance = 400.0`: 400m is a natural cutoff between pure sprint behavior and distances where endurance matters more.
- `sprint_reference_distance = 100.0`: 100m is the cleanest sprint reference because it directly represents raw short-distance speed.
- `reference_distance = 1500.0`: 1500m is a good middle anchor for a model intended to cover roughly 400m–5000m.
- `max_modeled_distance = 5000.0`: the current toy model is intentionally limited to this range.

Runner-Type Endurance Categories

The table `[running_model.runner_types]` defines reusable endurance exponents. These categories describe endurance quality only. They do *not* describe sprint speed.

Category	k	Interpretation
`elite`	1.05	Strong endurance. Performance degrades relatively slowly with distance.
`high`	1.06	Better than average endurance. Close to the classic Riegel ballpark for trained runners.
`average`	1.07	Middle-of-the-road endurance for this toy model.
`low`	1.09	Noticeably weak endurance. Times grow faster with distance.
`bad`	1.11	Poor endurance. Useful for athletes whose short-distance strength should fade rapidly as distance grows.

Why these values were chosen. The Riegel exponent is often near 1.06 for reasonably trained distance performance, so the current categories are centered near that value. The range 1.05 to 1.11 is broad enough to create visible separation between endurance profiles without making the toy model numerically unstable.

Per-Runner Attributes

Each runner entry combines the running model with the eating model.

`sprint_ref_time` Sprint reference time at `sprint_reference_distance`. Lower means faster sprint ability.

`ref_time` Endurance reference time at `reference_distance`. Lower means faster around 1500m.

`type` Which endurance category to use from `runner_types`.

`eat_base_sec` Base seconds for eating each hot dog.

`eat_slowdown_sec` Extra eating delay added per hot dog eaten.

`post_eat_linear` Linear increase in the running penalty after eating.

`post_eat_quadratic` Quadratic increase in the running penalty after eating.

`eat_noise_sec` Standard deviation of per-hot-dog eating noise.

`run_sprint_ref_time_rel_std` Relative noise scale on sprint reference time.

`run_ref_time_rel_std` Relative noise scale on endurance reference time.

`run_endurance_rel_std` Relative noise scale on the endurance exponent.

`post_eat_rel_std` Relative noise scale on the post-eating running multiplier.

Current Configured Athletes

Runner	Sprint Ref	1500m Ref	Type (long distance)	k
Chestnut	15.6s at 100m	390.0s	average	1.07
Bolt	9.7s at 100m	245.0s	high	1.06

Interpretation of the current values.

- Chestnut is configured as a slower runner but a much stronger eater.
- Bolt is configured as a much faster runner, both at sprint distance and around 1500m, but a weaker eater with stronger post-eating penalties.
- In the current TOML, Bolt uses the `high` endurance category. If the goal is a more sprint-specialist version of Bolt, his `type` can be moved to `low` or `bad` without any code change.

How to Read the Outputs

- The **probability plot** shows where athlete A wins often, rarely, or about half the time.
- The **deterministic plot** shows where the mean model favors one athlete over the other.
- The **tie curve** on the deterministic plot marks where the mean total times are equal.
- The printed **reference probabilities** are just spot checks from the config file.

Example Figure

Figure 1 shows a probabilistic phase diagram for Chestnut versus Bolt. The horizontal axis is the number of hot dogs eaten before the run, the vertical axis is the running distance, and the color gives $P(\text{Chestnut wins})$ under the Monte Carlo model.

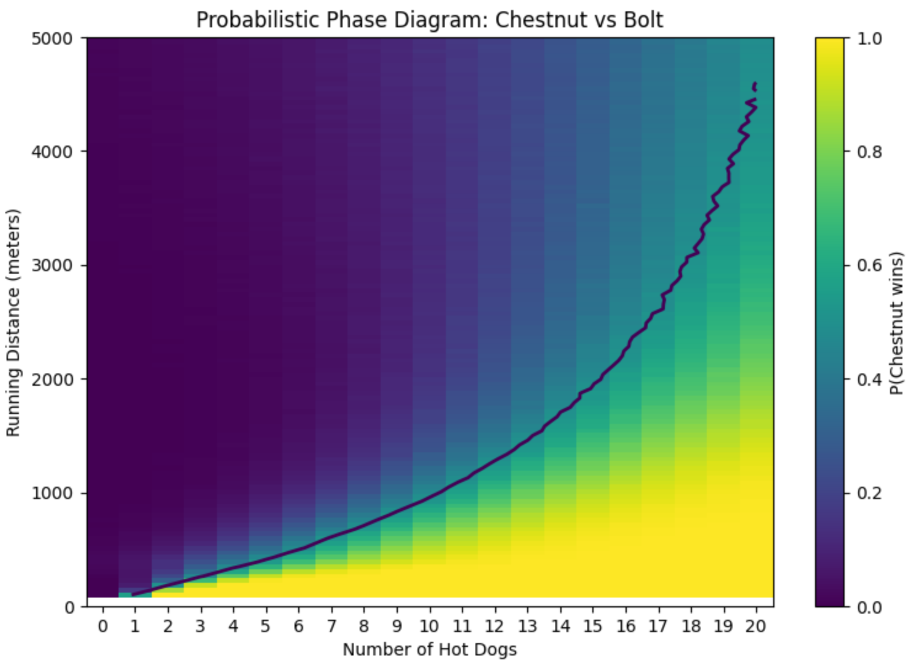

Figure 1: Probabilistic phase diagram for Chestnut versus Bolt. Yellow regions favor Chestnut, dark purple regions favor Bolt, and the curve marks the approximate 50-50 boundary.

The plot matches the intended story of the model. For short distances with enough hot dogs, Chestnut's eating advantage can outweigh Bolt's running advantage, so the lower part of the chart becomes bright. For longer distances, Bolt's stronger running parameters dominate, so the upper region stays dark unless the eating burden becomes very large. The boundary curve therefore slopes upward: as distance increases, Chestnut needs Bolt to be penalized by more hot dogs before the matchup becomes competitive.

Summary

The current project is a configurable toy simulator for "eat first, then run" matchups. Its main modeling choices are:

- a linear-plus-slowdown eating model,
- a multiplicative post-eating running penalty,
- a two-regime running model with a sprint segment up to 400m and a Riegel endurance segment beyond 400m,
- Monte Carlo noise around both eating and running parameters.

The TOML file controls all of the meaningful athlete behavior. Sprint strength is set by `sprint_ref_time`; endurance is set by `ref_time` and `type`/k; eating strength is set by the eating and post-eating parameters.